

Topic 7: AI for Technical Writing and Documentation

7.1 Code and API Documentation

- AI dramatically reduces the time spent on documentation -- one of the most resisted tasks in software development
- Generate from code: paste a function or class and ask 'Write complete documentation for this'
- README generation: 'Write a clear README for this project based on the code: [paste code and structure]'
- API reference: 'Document this endpoint: method, URL, path parameters, request body schema, example request, example response, error codes'
- Changelog entries: 'Write a changelog entry for these changes in the format: Added / Changed / Fixed / Removed'

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Documentation prompt:
'Generate documentation for this Python function. Include:
- One-sentence summary
- Detailed description of what it does
- Parameters table: name, type, description, required/optional
- Return value: type and description
- Example with realistic input and expected output
- Common errors and how to handle them

Function: [paste your code here]'
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7.2 User Guides and Manuals

- Audience-first: specify exactly who will read this -- new user or expert, technical or non-technical
- Task-based structure: organise by what users want to accomplish, not by how the product is built
- Step-by-step instructions: 'Write step-by-step instructions for [task], assuming the user has never done this before'
- Troubleshooting sections: 'Generate a troubleshooting guide for the 5 most common problems users face with [product]'
- Plain English check: 'Rewrite this section so a non-technical user can follow it without any background knowledge'

7.3 Process Documentation (SOPs and Runbooks)

- SOP format: 'Write an SOP for [process] including: purpose, scope, who is responsible, step-by-step procedure, exceptions'

- Runbooks: 'Write a runbook for deploying a new version of [system] including pre-checks, steps, rollback procedure'
- Process flowcharts: ask AI to write a text-based flowchart you can then recreate in Lucidchart, draw.io, or Miro
- Knowledge base from support tickets: 'Turn this support ticket thread into a reusable help centre article: [paste thread]'

7.4 Technical Report Writing

- Executive summary: 'Write a 1-page executive summary of this technical report for a non-technical CEO audience: [paste]'
 - Methodology sections: describe your approach in notes; ask AI to write it in formal, structured language
 - Data interpretation: paste tables or numerical results; ask AI for a written analysis of key findings and implications
 - Abstract generation: 'Write a 150-word abstract for this research paper that follows the Background / Methods / Results / Conclusion structure'
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